

SOLAR-POWERED AUTONOMOUS

WASTE SORTING ROBOT

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Declaration

We, the undersigned, declare that the project report **SUSTAINABLE BEACH CLEANUP USING SOLAR-POWERED AUTONOMOUS WASTE SORTING ROBOT** submitted for partial fulfillment of the requirements for the award of the degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala, is a bonafide work done by us under supervision of **Dr. BINDUMOL E.K.** This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the sources. We also declare we have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources that have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma, or similar title of any other University.

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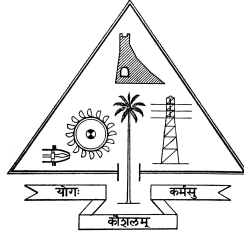
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Certificate

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Abstract

This project introduces a solar-powered autonomous robot specifically designed for efficient beach waste collection and sorting. Developed to address the growing issue of pollution along coastal areas, the robot leverages solar energy, making it an environmentally sustainable solution. The robot is equipped with a waste picking arm system operated by Servo Motors, allowing for efficient waste collection without human intervention and placing it in the intended bin. A Raspberry Pi and camera module enable real-time waste detection, an ESP32 guides the robot to collect visible waste materials. Additionally, the robot is fitted with an ultrasonic sensor for obstacle avoidance, which aids in smooth navigation across uneven beach terrains. The DC motor enables consistent, reliable movement, while the solar power system minimizes the need for external charging, making it ideal for remote and extensive beach areas. The advantages of this project are its autonomy, cost-effectiveness, and minimal energy dependency, positioning it as a robust solution for beach cleanup operations. By reducing manual labor and maintaining a cleaner shoreline, this beach waste-collecting-sorting robot promotes environmental health and sustainable practices, offering significant potential for broader environmental management applications.

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Vision of the Institution

To be a Premier Institution of Excellence in Engineering Education and Research for Sustainable Development.
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Mission of the Institution

1.	Provide Quality Education in Engineering and Technology.
2.	Foster Passion for Research.
3.	Transform the Students into Committed Technical Personnel for the Social and Economic Wellbeing of the Nation.

Vision of the Department

Be a centre of excellence in Electrical and Electronics Engineering having strong learning and research environment, capable of making significant contributions for sustainable development.

Mission of the Department

To generate world-class engineers to meet challenges in the field of Electrical and Electronics Engineering through innovations and pursuit of new knowledge and make them ethically sound and professionally competent to provide service to the society.
--

Program Educational Objectives (PEO)

1.	To mold the students with global standards by providing knowledge in Basic sciences and Mathematics in general and Electrical & Electronics Engineering in particular.
2.	To impart necessary skill and practical knowledge to design, analyze and synthesize Electrical & Electronic systems using modern tools.
3.	To equip students with communication skills, professional ethics, and teamwork for a successful career.
4.	To develop leadership qualities, innovation, and lifelong learning skills among students.

Program Outcomes (POs)

PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop solutions to complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions.
PO3	Design/Development of Solutions: Design solutions for complex engineering problems and design system components/processes to meet desired needs.
PO4	Conduct Investigations of Complex Problems: Conduct investigations using research-based knowledge including design of experiments and interpretation of data.
PO5	Modern Tool Usage: Select and apply appropriate techniques, resources and modern engineering tools for complex engineering activities.
PO6	The Engineer and The Society: Analyze and evaluate societal, health, safety, legal and cultural issues relevant to engineering practice.
PO7	Environment and Sustainability: Understand impact of engineering solutions in environmental contexts and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities.
PO9	Individual and Team Work: Function effectively as an individual and as a member or leader in teams.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and society.
PO11	Project Management and Finance: Apply engineering and management principles to manage projects.
PO12	Life-long Learning: Recognize the need for lifelong learning and adapt to technological change.

Program Specific Outcomes (PSOs)

PSO1	Intelligent Embedded System Design: Ability to design and develop an integrated embedded system using Raspberry Pi, ESP32, and sensors for real-time applications.
PSO2	AI-based Object Detection Implementation: Ability to implement and deploy YOLO-based computer vision models for real-time waste detection and classification.
PSO3	Autonomous Robotic System Development: Ability to build and control an autonomous robotic system capable of navigation, obstacle avoidance, and task execution.
PSO4	Renewable Energy Integration: Ability to design systems powered by solar energy with efficient battery and power management for sustainable operation.
PSO5	Mechatronics and Actuation Systems: Ability to integrate motors, servo mechanisms, and mechanical subsystems for precise movement, picking, and sorting actions.
PSO6	IoT-based Monitoring and Control: Ability to develop IoT-enabled solutions using ESP32 and mobile applications for remote monitoring and control.
PSO7	System Integration and Optimization: Ability to integrate hardware and software components into a cohesive system ensuring reliability, efficiency, and real-time performance.
PSO8	Environmental and Societal Impact Awareness: Ability to apply engineering solutions toward sustainable waste management and environmental conservation.

Sustainable Development Goals (SDGs) Addressed

SDG 6	Clean Water and Sanitation: Helps reduce beach pollution, preventing waste from entering water bodies and improving water quality.
SDG 7	Affordable and Clean Energy: Utilizes solar energy for operation, promoting renewable and sustainable power usage.
SDG 11	Sustainable Cities and Communities: Contributes to cleaner public spaces by enabling automated waste collection in coastal and urban areas.
SDG 12	Responsible Consumption and Production: Supports waste segregation and proper disposal, encouraging recycling and responsible waste management.
SDG 13	Climate Action: Reduces environmental impact by promoting eco-friendly waste management and minimizing pollution.
SDG 14	Life Below Water: Prevents marine pollution by removing waste from beaches, protecting marine ecosystems and biodiversity.

SDGs addressed with Justification

Project Title	SDG addressed	Justification
Sustainable Beach Clean Up Using Solar Powered Autonomous Waste Sorting Robot	SDG 6 - Clean Water and Sanitation	The system removes waste from beaches before it reaches water bodies, thereby reducing water contamination and improving coastal sanitation.
	SDG 7 - Affordable and Clean Energy	By integrating a solar-powered energy system, the project minimizes dependence on non-renewable sources and promotes sustainable energy usage.
	SDG 11 - Sustainable Cities and Communities	The solution supports cleaner public environments by automating waste collection in beaches and can be extended to urban spaces, enhancing livability.
	SDG 12 - Responsible Consumption and Production	Through waste detection and segregation, the system encourages recycling and proper disposal, contributing to efficient resource management.
	SDG 13 - Climate Action	The use of renewable energy and reduction of unmanaged waste helps lower environmental impact and supports climate-conscious practices.
	SDG 14 - Life Below Water	By preventing beach waste from entering oceans, the system helps protect marine life and reduce ocean pollution.

POs & PSOs addressed with Justification

Project Title	POs & PSOs	Justification
Sustainable Beach Clean Up Using Solar Powered Autonomous Waste Sorting Robot	PO1	Applied knowledge of embedded systems, electronics, and control to develop an integrated system using Raspberry Pi, ESP32, and sensors.
	PO2	Identified key challenges such as waste detection, autonomous navigation, obstacle avoidance, and efficient waste collection in beach environments.
	PO3	Designed a complete system integrating solar power, AI-based detection (YOLOv8), robotic arm, and sorting mechanism.
	PO5 & PSO2	Utilized modern tools such as YOLOv8, Raspberry Pi, OpenCV (support), and IoT platforms for real-time implementation.
	P06	Developed a solution addressing environmental issues like beach pollution and waste management.
	P07 & PS03	Integrated solar power and efficient energy management to promote sustainable and eco-friendly operation.
	PS04	Developed an autonomous robotic system capable of navigation, waste collection, and sorting.
	PS05	Integrated multiple subsystems (vision, control, power, actuation) into a unified working prototype.

Chapter 1

Introduction

Oceans, covering approximately 70% of the Earth's surface, play a crucial role in regulating global temperatures, sequestering atmospheric carbon dioxide, and supporting the livelihoods of over a billion people. However, ocean pollution has emerged as a global concern, with an estimated 75–199 million tons of plastic waste already accumulated and an additional 33 billion pounds entering annually. This pollution severely impacts marine ecosystems, biodiversity and coastal health, resulting in dead zones and the loss of millions of marine species each year. The primary causes include ineffective waste management and land-based nonpoint source pollution.

In Kerala, coastal pollution control measures currently involve the use of a 39 lakh diesel-powered beach cleaning machine. Although effective in waste removal, it's fossil fuel dependency contributes to emissions and environmental degradation. To address this challenge, an IoT-enabled autonomous solar-powered beach cleaning robot has been developed as an eco-friendly alternative. The robot employs advanced image processing algorithms to detect, collect and sort waste efficiently while operating on renewable solar energy.

This sustainable system minimizes fossil fuel use, reduces manual labor requirements, and aligns with global initiatives aimed at protecting marine life and reducing coastal pollution. It represents a significant step toward achieving cleaner shorelines, sustainable coastal management and a reduced environmental footprint in waste collection operations.

1.1 Problem Statement

Inefficient Waste Detection: Existing robotic systems struggle to accurately identify and differentiate waste from natural elements like sand, shells, and seaweed, leading to false detections and missed debris.

Use of Non-Renewable Energy for Beach Cleaning: Traditional beach cleaning machines rely on fuel-powered tractors, leading to high costs, carbon emissions, and sustainability concerns. This emphasizes the need for an eco-friendly and cost-effective alternative.

Limited Autonomous Waste Collection: Many existing robots rely on manual remote control, whereas an autonomous navigation system is needed for better efficiency.

1.2 Objectives

- To model a solar PV based waste picking arm system to navigate, clean and gather various types of visible waste on the beach.
- To detect types of wastes in beach using Raspberry Pi based camera module and ensure targeted collection.
- To model a system to sort the collected waste and deposit to the intended bin.

Chapter 2

Literature Survey

The development of autonomous beach cleaning robots has gained traction as researchers aim to address environmental pollution with innovative and sustainable solutions. Several notable projects highlight the advancements in design, automation, and operational efficiency of these robots.

“Design and Construction of Beach Cleaning Trailer by Finite Element Method,” by Kuson Prakobkarn, Banyat Saitthiti, Sakda Intaravichi, which uses a vibrating mesh-based beach cleaning robot pushes sand into a chain conveyor that transports it to a shaking tray for waste separation. Trashes are collected in a trash tray, but the process inadvertently collects seashells and living organisms like crabs, leading to their destruction [1].

In “Binman: An Autonomous Beach Cleaning Robot,” D. Varghese and A.Mohan (2022) introduced “Binman,” a robot equipped with advanced autonomous navigation and object detection capabilities specifically tailored for beach cleaning. Utilizing a network of sensors and intelligent algorithms, Binman can autonomously plan and execute cleaning routes while avoiding obstacles on uneven, sandy terrain. This study discusses the challenges of maintaining stability on such surfaces and optimizing the robot’s path for efficient area coverage [2].

The “Design of Beach Cleaning Robot (BeBot)” by F. A. Alkaabi et al.(2023) presents BeBot, a modular and environmentally friendly robot optimized for sustainable beach cleaning. The design prioritizes low environmental impact, energy efficiency, and operational versatility, enabling it to adapt to various beach types. BeBot’s integration of smart navigation systems allows it to effectively locate and collect waste, with a focus on ensuring minimal disruption to the beach ecosystem. This research emphasizes the importance of sustainability and adaptability in robotic beach cleaning solutions [3].

Development of an autonomous beach cleaning robot “Hirottaro” by Ichimura, and S. I. Nakajima, has a better residual collecting mechanism which even can collect plastic pieces and cigarette butts but the efficiency of the waste collecting mechanism is doubted as the mechanism have a complex structure to move on [4].

The “Design and Fabrication of Beach Sand Cleaning Machine” by V. Mepani et al. (2020) offers a more accessible, community-oriented approach. This manually operated machine uses a screening mechanism to sift debris from sand and is built to be simple and cost-effective. Although it lacks full automation, the design is suitable for smaller beach areas, emphasizing ease of use and affordability. This study shows how low-cost beach-cleaning solutions can effectively engage local communities in environmental cleanup efforts [5].

The RF Controlled Beach Cleaner project features a four-wheel drive vehicle equipped with a cleaning mechanism and a dirt bag for garbage collection. This robot operates using a chain and sprocket system powered by three electrical motors: two for locomotion and one for the cleaning mechanism. A microcontroller processes directional commands received via RF remote control, enabling automatic trash collection while separating sand from debris using a wire mesh [6].

Finally, “AI-Based Robot for Beach Cleaning,” by Ashitha V Naik (2023) explores the integration of artificial intelligence in robotic beach cleaning. The robot utilizes machine learning algorithms for real-time waste detection and classification, focusing particularly on identifying non-biodegradable materials such as plastic. With image recognition capabilities, this AI-driven robot can prioritize the collection of high-risk pollutants, showing the value of AI for enhanced environmental awareness and precision in waste sorting [7].

Chapter 3

Design and Methodology

Solar-powered, autonomous beach cleaning and waste-sorting robot is designed to address the growing issue of coastal pollution through sustainable technology. The robot integrates IoT-based control, image processing for waste detection, and a servo-driven collection mechanism to identify and gather litter from beach surfaces efficiently. Collected waste is then automatically sorted into separate compartments for plastics, metals, and biodegradable materials, enabling effective recycling and disposal. Powered by renewable solar energy and equipped with ultrasonic sensors for obstacle avoidance, the system ensures safe, energy-efficient, and eco-friendly operation. This design provides a modern, sustainable alternative to conventional fossil fuel based cleaning methods.

3.1 System Architecture

The block diagram given in the Figure 3.1 represents the architecture of the solar-powered autonomous beach-cleaning robot, showing the relationship between its components and how they work together to perform beach waste collection and sorting.

(i) Power Supply System

- The solar panel shown in Figure 3.2 is a 3W, 9V photovoltaic module designed for low-power, off-grid applications. It harnesses solar energy and converts it into electrical energy, which can either be used directly or stored in a battery for later use. In the proposed system, the generated solar power is utilized to support the robot's operation. The output from the solar panel is first supplied to a charge controller, which regulates the charging process. This ensures efficient power management and protects the battery from

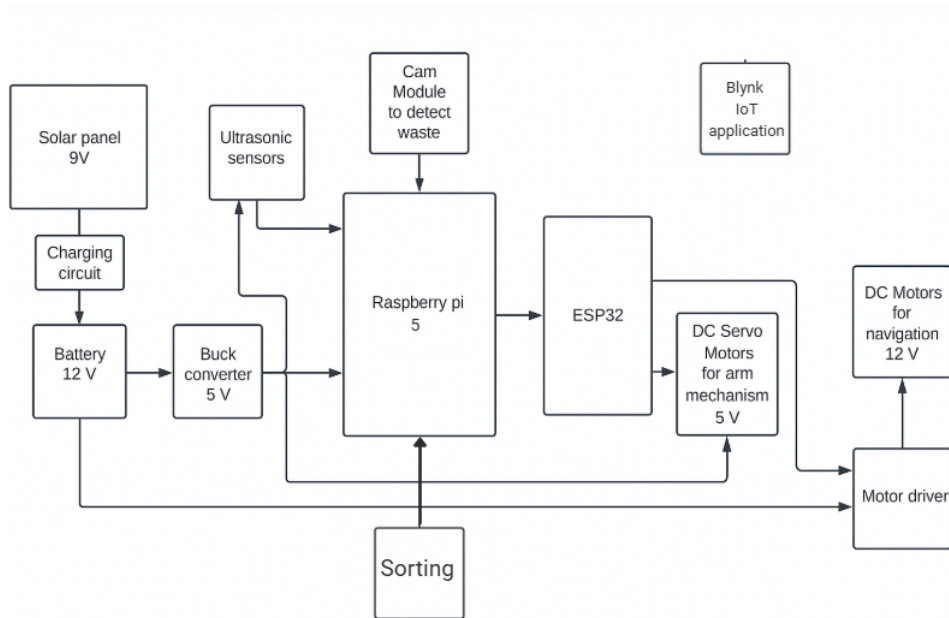


Figure 3.1: Block Diagram of Robot

overcharging, thereby enhancing the reliability and lifespan of the power system.

- The charge controller then charges the battery, which serves as an energy storage system, providing a steady power supply to the robot, even in cloudy conditions or during nighttime. The battery supplies power to the Raspberry Pi- 5 and other components, enabling continuous operation.

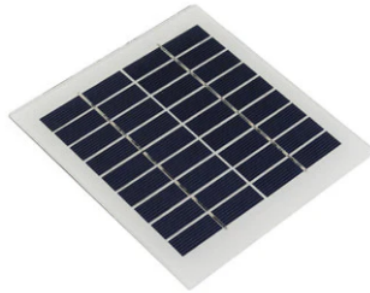


Figure 3.2: 3W Solar Panel

12V Battery

- The 12V battery shown in the Figure 3.3 powers the motors, servo mechanisms, and other essential components of the waste-collecting robot. It provides a stable power supply for smooth navigation and efficient waste collection. The battery is rechargeable through a 3W solar panel, ensuring continuous operation and sustainability. It delivers consistent output, preventing power fluctuations that could affect performance. Designed for extended use, it enhances the robot's reliability in outdoor environments. By integrating solar charging, the system reduces dependence on external power sources, making it an eco-friendly solution for autonomous waste collection and sorting.

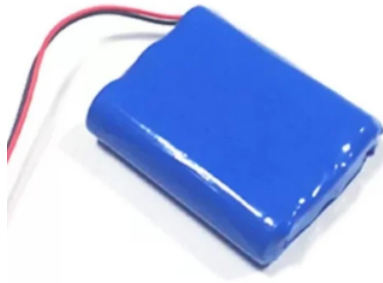


Figure 3.3: 12V Battery

(ii) Central Processing Unit (Raspberry Pi- 5)

- The Raspberry Pi- 5 shown in the Figure 3.4 acts as the central control unit of the robot, processing data from various sensors and controlling different motors. It receives power from the battery and coordinates the robot's operations based on the inputs from connected sensors and modules. There is a serial communication between Raspberry Pi-5 and ESP-32 through which the data about the waste detection is transferred.



Figure 3.4: Raspberry Pi 5

(iii) **ESP32**

- The ESP32 shown in the Figure 3.5 controls key functions in the robot, including waste-picking operations, autonomous movement, IoT connectivity, and communication with the Raspberry Pi. It manages the servo motors for efficient waste collection and sorting, processes data from ultrasonic sensors for obstacle avoidance, and enables remote monitoring through Wi-Fi. Acting as a communication bridge, the ESP32 exchanges data with the Raspberry Pi, which provides real-time waste detection inputs, ensuring smooth coordination and effective operation of the robot.

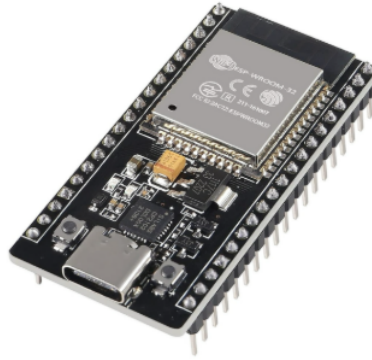


Figure 3.5: ESP 32

(iv) **Camera Module**

- The camera module shown in the Figure 3.6 is connected to the Raspberry Pi and is responsible for capturing images of the beach surface. Using image processing algorithms, the Raspberry Pi analyses the images from the camera to detect visible waste on the beach, guiding the robot's waste collection system when trash is spotted.

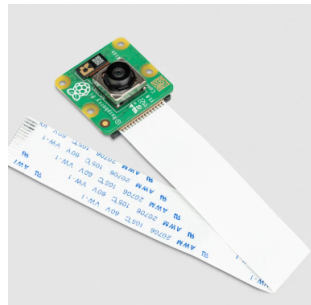


Figure 3.6: Raspberry Pi Camera Module

(v) **Ultrasonic Sensor**

- The ultrasonic sensor shown in the Figure 3.7 is also connected to the ESP 32. It detects obstacles in the robot's path by emitting ultrasonic waves and measuring the reflected waves. When an obstacle is detected within a certain range, the Raspberry Pi adjusts the robot's direction to avoid collisions, allowing it to navigate safely across the beach.



Figure 3.7: Ultrasonic Sensor

(vi) **Motor Controller**

- The robot is equipped with a dual motor controller shown in the Figure 3.8, which are controlled by the ESP-32. These controllers regulate the power and signals sent to the motors, enabling precise control over movement and operation.

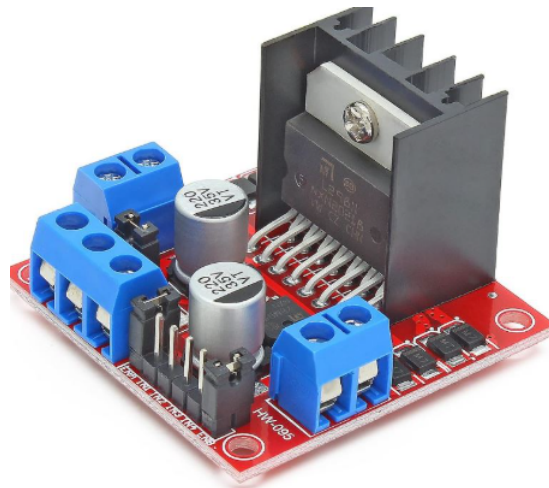


Figure 3.8: L298N

(vii) **DC Motor**

- The DC motor shown in the Figure 3.9 drives the wheels of the robot, enabling it to move along the beach. Controlled by the motor controller, the DC motor receives commands from the ESP-32 to adjust speed and direction as needed, based on the robot's path and any obstacles detected by the ultrasonic sensor.



Figure 3.9: Johnson Geared DC Motor (100 RPM)

(viii) **Servo-Motors**

- The MG996R shown in the Figure 3.10 is a popular high-torque servo motor with metal gears, offering durability and reliability. It operates on 4.8V–7.2V, delivering a torque of up to 11kg/cm at 6V. With a speed of 0.17s/60°, it provides precise movement control. The waste-picking arm system is powered by servo motors operates. The system typically consists of a mechanical arm with multiple joints, each controlled by servo motors.. Its internal circuitry includes a potentiometer for accurate positioning and a control signal response. The MG996R is favored for its affordability, ease of use, and ability to handle moderate loads, making it suitable for projects requiring strength, stability, and precise rotational movement.



Figure 3.10: MG996R Servo Motor

(ix) **Buck Converter**

- A buck converter shown in the Figure 3.11 efficiently steps down 12V to 5V using high-frequency switching, an inductor, a diode, and a capacitor. It provides stable power for the servo motor and ultrasonic sensor with high efficiency, reducing heat and energy loss. This ensures longer battery life and reliable robot operation



Figure 3.11: Buck Converter - LM2596

3.2 Flowchart of Robot



Figure 3.12: Flowchart of Robot

The flowchart in Figure 3.12 represents the working logic of the beach waste collecting robot using Raspberry Pi and IoT-based control.

1. **Start** : The process begins with the system initialization.
2. **Initialization** : The robot initializes sensors, actuators, and the IoT mobile application (Blynk). This ensures all components are ready for operation.

3. **Reading Input from Mobile App** :The system reads virtual button inputs from Blynk, which allows manual or automatic control.
4. **Mode Selection** : If the Mode Select is ON, the robot operates autonomously. If the Mode Select is OFF, manual control is enabled, allowing the user to navigate the robot using the mobile application (forward, backward, left, right).
5. **Autonomous Mode (If Mode is ON)** : The system checks if it has received serial data from Raspberry Pi, which likely indicates the detection of a waste.
 - If data is received - The servo motor is activated to pick up the waste.
 - If no data is received - The robot proceeds to obstacle detection using ultrasonic sensors.
6. **Obstacle Detection** : The robot reads values from ultrasonic sensors. If an obstacle is detected, the robot turns right to avoid it and continues scanning.
7. **Stop Condition** : The process continues in a loop until the system is manually stopped.

3.3 Flowchart of Sorting Mechanism

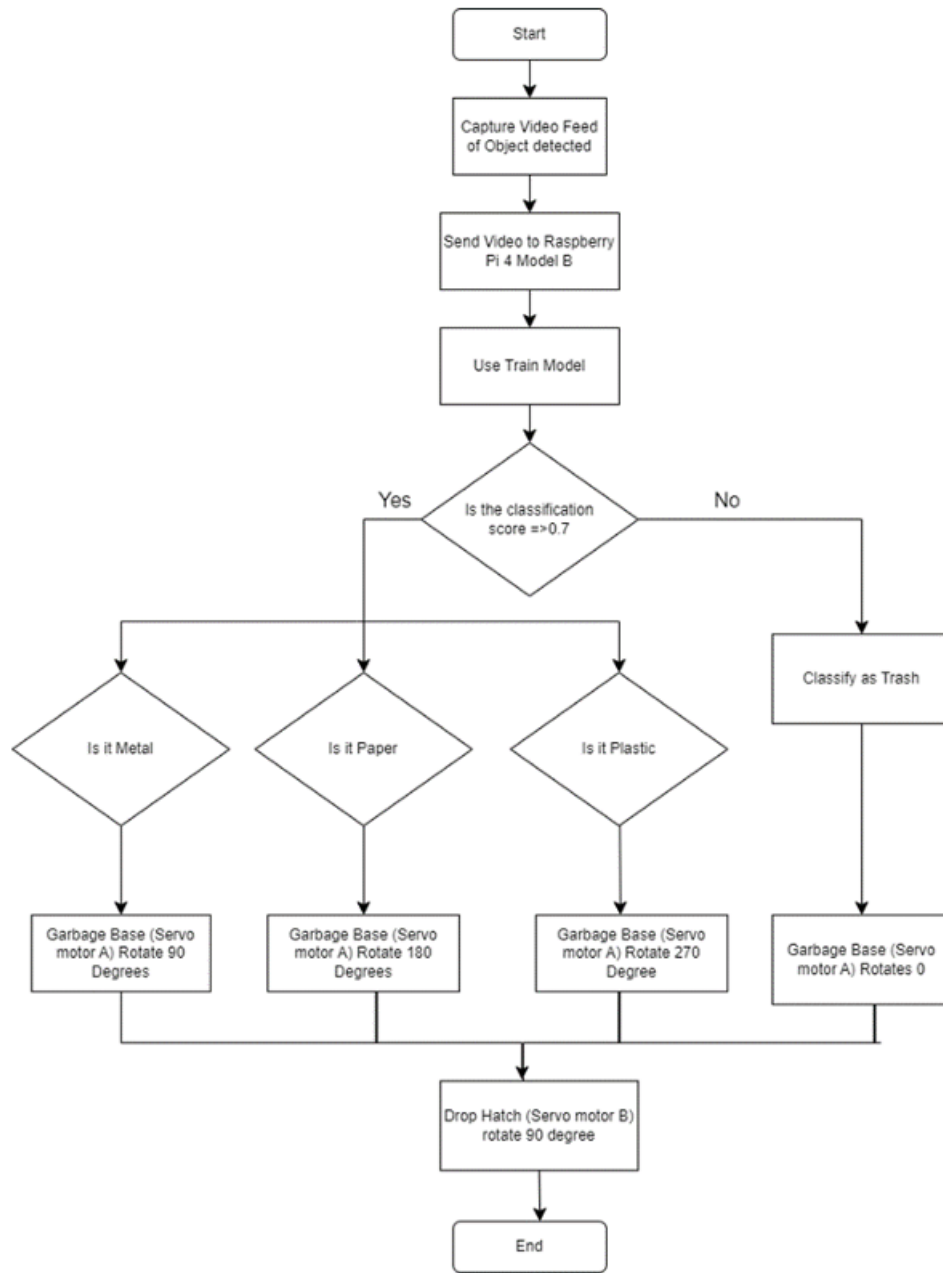


Figure 3.13: Flowchart of Sorting Mechanism

The flowchart in figure 3.13 represents the sorting logic of the beach robot using Raspberry Pi.

1. **Start:** The system powers on and the robot enters its waste-detection routine.
2. **Capture video feed of detected object:** The onboard camera continuously scans the ground ahead. When a candidate object (potential waste) is detected

in the camera frame, a short video clip or image sequence of the object is captured for processing.

3. **Send video to Raspberry Pi 5 Model B:** The captured video frames are transferred to the Raspberry Pi 5 (the primary processing unit) where the image-processing pipeline and classifier run.
4. **Run trained model on the input:** The Raspberry Pi processes the incoming frames through the pre-trained classification model (object detector + classifier) to identify the object type.
5. **Confidence Evaluation:** The model's classification confidence is checked. If the score is below 0.7, the object is labeled as trash and sent to the general waste bin.
6. **Material Identification:** If the confidence is ≥ 0.7 , the system determines the specific waste category metal, paper, or plastic based on the model output.
7. **Servo Motor Alignment:** The base servo motor (Servo A) rotates to a predefined angle (0° , 90° , 180° , or 270°) corresponding to the correct waste compartment.
8. **Waste Deposition:** The drop hatch (Servo B) activates, rotating 90° to release the collected waste into the selected compartment of the sorting bin.
9. **Reset and Continue Operation:** After deposition, the system resets the servo positions and resumes scanning for the next object, continuing the autonomous cleaning process.

3.4 Kinematic Equations

The design and development of the proposed autonomous beach cleaning robot require a systematic evaluation of its kinematic parameters, power requirements, and mechanical performance. This section presents the fundamental calculations that govern the robot's motion, load handling capability, and energy consumption.

Initially, the power system design is analyzed to ensure that the robot can operate efficiently using a combination of solar energy and battery storage. This includes estimating the total energy requirement, battery capacity, and the contribution of the solar panel to extend operational time.

Further, the torque calculations are carried out to determine the minimum torque required by the motors to move the robot under load conditions, especially considering the high rolling resistance of sandy terrain. These calculations ensure appropriate motor selection for reliable operation.

The section also evaluates the relationship between linear velocity and wheel rotational speed (RPM) to determine suitable motor speed characteristics for achieving the desired movement of the robot.

3.4.1 Design of Solar Power and Battery

For 1-hour operation:

$$E_{\text{bat}} = P_{\text{total}} \times t = 29 \times 1 = 29 \text{ Wh} \quad (3.1)$$

Battery capacity required:

$$C_{\text{bat}} = \frac{E_{\text{bat}}}{V_{\text{bat}}} = \frac{29}{12} = 2.42 \text{ Ah} \quad (3.2)$$

Using a 12 V, 3 Ah battery ensures safe operation for 1 hour.

Solar panel details:

$$P_{\text{solar}} = 3 \text{ W}$$

$$t_{\text{sun}} = 5 \text{ h}$$

$$\eta = 0.8$$

Solar energy generated:

$$E_{\text{solar}} = P_{\text{solar}} \times t_{\text{sun}} \times \eta \quad (3.3)$$

$$E_{\text{solar}} = 3 \times 5 \times 0.8 = 12 \text{ Wh} \quad (3.4)$$

Operation time:

$$t_{\text{without PV}} = \frac{E_{\text{bat}}}{P_{\text{total}}} = \frac{29}{29} = 1.0 \text{ h} \quad (3.5)$$

$$t_{\text{with PV}} = \frac{E_{\text{bat}} + E_{\text{solar}}}{P_{\text{total}}} = \frac{29 + 12}{29} = 1.41 \text{ h} \approx 1 \text{ h } 25 \text{ min} \quad (3.6)$$

3.4.2 Torque Calculation for 7 kg Load

Given:

$$W = 7 \text{ kg}$$

$$\mu = 0.35 \quad (\text{Coefficient of rolling resistance for sand})$$

$$g = 9.81 \text{ m/s}^2$$

$$r = 0.05 \text{ m}$$

$$n_{\text{wheel}} = 2$$

Total rolling resistance force:

$$F_{\text{total}} = \mu \times W \times g \quad (3.7)$$

$$F_{\text{total}} = 0.35 \times 7 \times 9.81 = 24.06 \text{ N} \quad (3.8)$$

Force per wheel:

$$F_{\text{wheel}} = \frac{F_{\text{total}}}{n_{\text{wheel}}} = \frac{24.06}{2} = 12.03 \text{ N} \quad (3.9)$$

Torque required per motor:

$$\tau = F_{\text{wheel}} \times r = 12.03 \times 0.05 = 0.6015 \text{ Nm} \quad (3.10)$$

Converting to kg-cm:

$$\tau = 0.6015 \times 10.2 = 6.14 \text{ kg-cm} \quad (3.11)$$

Hence, each motor should provide at least 6.14 kg-cm torque.

3.4.3 Desired Linear Speed and Corresponding RPM

Desired linear speed:

$$v = 1 \text{ m/s}$$

Relationship between wheel speed and linear velocity:

$$v = 2\pi r \times \frac{\text{RPM}}{60} \quad (3.12)$$

Rearranging:

$$\text{RPM} = \frac{v \times 60}{2\pi r} \quad (3.13)$$

Substituting $v = 1 \text{ m/s}$, $r = 0.05 \text{ m}$

$$\text{RPM} = \frac{1 \times 60}{2\pi \times 0.05} = \frac{60}{0.314} \approx 191 \text{ RPM} \quad (3.14)$$

Thus, a 12 V DC motor with approximately 200 RPM is suitable.

3.4.4 Power Calculation

Given:

$$V_{\text{DC}} = 12 \text{ V}$$

$$I_{\text{DC}} = 0.5 \text{ A per motor}$$

$$n_{\text{DC}} = 2$$

$$V_{\text{servo}} = 5 \text{ V}$$

$$I_{\text{servo}} = 0.6 \text{ A per servo}$$

$$n_{\text{servo}} = 5$$

$$P_{\text{misc}} = 2 \text{ W}$$

Power for DC motors:

$$P_{\text{DC}} = n_{\text{DC}} \times V_{\text{DC}} \times I_{\text{DC}} \quad (3.15)$$

$$P_{\text{DC}} = 2 \times 12 \times 0.5 = 12 \text{ W} \quad (3.16)$$

Power for servo motors:

$$P_{\text{servo}} = n_{\text{servo}} \times V_{\text{servo}} \times I_{\text{servo}} \quad (3.17)$$

$$P_{\text{servo}} = 5 \times 5 \times 0.6 = 15 \text{ W} \quad (3.18)$$

Total power consumption:

$$P_{\text{total}} = P_{\text{DC}} + P_{\text{servo}} + P_{\text{misc}} \quad (3.19)$$

$$P_{\text{total}} = 12 + 15 + 2 = 29 \text{ W} \quad (3.20)$$

Parameter	Value
Required torque per motor	6.14 kg-cm
Required speed	191 RPM
Total power consumption	29 W
Recommended battery	12 V, 3 Ah
Operation time before solar	1 hour
Operation time after solar	1 hour 25 minutes

Table 3.1: Summary of Mechanical and Power Parameters

Component	Input Voltage	Output Voltage	Max Current	Other Specs
DC Motor Driver L298N	46V	5 -35V	2A/channel	-
DC Motor (Johnson)	12V	-	1.6A	100 RPM, 10 Nm
Battery	-	12V	13.5A	Li-ion 3S, 4500mAh
Raspberry Pi 5 Model B	5.1V	-	5A	-
ESP32 NodeMCU 32S	3 - 3.6V	-	500mA	-
Servo Motor MG996R	4.8-7.2V	-	2.5A	-
LM2596 Buck	3-40V	1.5-35V	3A	-
XY3606 Buck	24/12V	5V	5A	-
XL4015 Buck	5-36V	1.25 -32V	5A	-
Stepper Motor 28YBJ-48	5V	-	500mA	-
Stepper Driver ULN2003	5-12V	50V	500mA	-
Solar Panel	9V, 3W	11V	300mA	1000 W/m ²

Table 3.2: System Components Specifications

Chapter 4

Object Detection Algorithm

This chapter consists of the implementation of the waste detection algorithm.

4.1 Image Detection with Raspberry Pi, YOLOv8n

- The Raspberry Pi acts as the main processing unit, handling real-time image input from the camera module.
- A Raspberry Pi Camera Module captures continuous video frames for analysis.
- A YOLOv8n (You Only Look Once) deep learning model is deployed for fast and efficient object detection.
- YOLOv8n performs end-to-end detection, directly identifying waste objects without the need for separate preprocessing pipelines.
- The system enables real-time detection and classification of waste such as plastic, metal, and paper.

4.2 Algorithm for Raspberry Pi based Image Detection

This section outlines the algorithm used for real-time object detection using a Raspberry Pi and YOLOv8n. The model detects objects from a live camera feed and highlights detected objects.

- **Step 1: Initialization**

1. Import required libraries such as Ultralytics YOLO, OpenCV, NumPy, and OS.

2. Load the trained YOLOv8n model weights.
3. Define class labels (e.g., plastic, metal, paper).
4. Initialize the camera module for real-time video capture.
5. Set detection parameters such as confidence threshold.

- **Step 2: Video Capture and Preprocessing**

1. Start a continuous loop to process video frames.
2. Capture an image frame from the video stream.
3. Resize or format the frame if required by the model.
4. Pass the frame directly to the YOLOv8n model for detection.

- **Step 3: Model Inference**

1. The YOLOv8n model processes the frame in a single pass.
2. It outputs:
 - Bounding box coordinates.
 - Class labels.
 - Confidence scores.

- **Step 4: Object Detection and Visualization**

1. Iterate through the detected objects.
2. If the confidence score is above the threshold (0.5):
 - Retrieve the detected object's class label.
 - Check if the detected object is a plastic, metal or paper.
 - Convert normalized coordinates to actual pixel values.
 - Draw a bounding box around the detected object.
 - Display a label with the detected class name.
 - Print the detected object name to the console.

- **Step 5: Display and Cleanup**

1. Display the processed frame with bounding boxes and labels.
2. Continue capturing and processing frames until the ESC key is pressed.
3. Release camera resources and close all display windows.

Chapter 5

Results and Discussion

The developed beach cleaning robot integrates multiple electronic and mechanical subsystems to achieve autonomous navigation, waste detection, and collection. The system operation is based on coordinated interaction between the control, sensing, actuation, and power modules.

The robot is powered by a battery-supported system, supplemented by a solar panel, ensuring continuous operation. The ESP32 functions as the central control unit, managing motor actuation and processing inputs from sensors. The Raspberry Pi 5 serves as the primary processing unit for vision-based tasks.

For mobility, the robot is equipped with 12V high-torque DC motors connected to the wheels and driven through an L298N motor driver. Based on control signals from the ESP32, the robot is capable of moving forward, backward, and turning, enabling effective navigation on sandy terrain.

The working of the system begins with continuous environment monitoring using the camera module. The Raspberry Pi processes the captured images using a YOLOv8n-based object detection algorithm to identify waste materials such as plastic, paper, or metal. Simultaneously, an ultrasonic sensor measures the distance to nearby obstacles, allowing the robot to avoid collisions and maintain smooth navigation.

Once waste is detected, the Raspberry Pi communicates the detection result to the ESP32, which then controls the actuation mechanism. The robot navigates toward the detected object, and a servo motor (MG996R) activates the robotic arm to pick up the waste. The collected waste is then deposited into an onboard bin. A stepper motor mechanism facilitates bin positioning or sorting operations when required.

The entire process follows a continuous cycle of detection - navigation - obstacle - avoidance - collection - sorting, enabling autonomous operation. The chassis structure, combined with high-torque motors and appropriate wheel design, ensures stability and mobility across uneven beach surfaces.

5.1 App Based Control Interface

The developed system includes an IoT-based control interface shown in the Figure 5.1 implemented using a mobile application, enabling real-time interaction with the robot. The interface was designed to provide both manual control and autonomous operation modes, ensuring flexibility and ease of use.

The control panel consists of multiple virtual buttons for directional movement, task execution, and system monitoring. The FORWARD, BACKWARD, LEFT, and RIGHT controls allow precise navigation of the robot during manual operation. These controls were tested and demonstrated smooth and responsive movement of the robot on sandy terrain.

A central status indicator is incorporated to display the operational state of the robot (ON/OFF), ensuring clear system feedback to the user. Additionally, dedicated controls such as “PICK” and “SORT” enable manual triggering of the robotic arm for waste collection and bin sorting operations.

The interface also includes a mode selection feature, allowing the user to switch between manual mode and autonomous mode. In autonomous mode, the robot performs waste detection, navigation, collection, and sorting without human intervention. In manual mode, the user can directly control all movements and actions through the application.

The communication between the mobile application and the robot is established via Wi-Fi using the ESP32 module, ensuring real-time data exchange and low-latency control. During testing, the interface demonstrated reliable connectivity, quick response time, and effective synchronization with the robot’s hardware components.

5.2 Prototype Implementation

The developed system was successfully implemented as a solar-powered autonomous beach cleaning robot capable of waste detection, collection, and sorting. The integration of hardware and software components resulted in a functional prototype demonstrating the intended objectives. The YOLOv8n-based object detection model deployed on the Raspberry Pi enabled real-time detection of waste materials such as plastic, paper, and metal with satisfactory accuracy under controlled conditions. The robot demonstrated autonomous navigation, detecting waste and moving toward the target for collection. The ultrasonic sensor provided effective obstacle detection, reducing collision risks during operation. The servo-based robotic arm (MG996R) successfully performed picking operations, while the stepper motor mechanism enabled waste sorting into designated bins. The ESP32-based IoT interface allowed real-time monitoring, manual control, and switching between

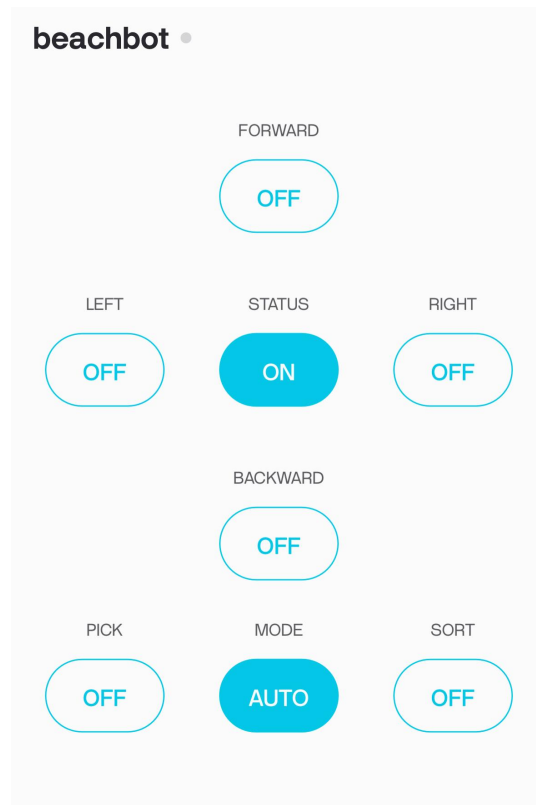


Figure 5.1: User Interface



Figure 5.2: Side view of the prototype

manual and autonomous modes. The solar-powered system with a 3S Li-ion battery supported continuous operation, validating the feasibility of renewable energy integration. The system showed coordinated functioning of detection, control, and actuation modules during testing.



Figure 5.3: Front view of the prototype

During the development and testing phases, several challenges were identified. The YOLOv8n model showed reduced accuracy under poor lighting conditions, shadows, and partially occluded waste objects. The Raspberry Pi experienced limited processing capability, leading to occasional delays in detection and response time. Movement on sandy terrain caused wheel slippage and reduced stability, affecting precise navigation.

Chapter 6

Conclusion

The Solar-Powered Autonomous Waste Sorting Robot for Sustainable Beach Cleanup is an innovative solution to address beach pollution by integrating solar power, autonomous navigation, IoT-based control, and YOLO based image detection using Raspberry Pi. The robot collects waste using its robotic arm and sorts it into bins, minimizing manual labor while improving efficiency and reducing costs. A battery backup system ensures continuous operation even in low sunlight conditions, enhancing its reliability. The robot features solar-powered operation, ultrasonic sensors for navigation, and an IoT-based control system for real-time tracking. YOLO based image detection enhances waste identification, while a ESP 32 processes data for seamless operation. Despite its effectiveness, the system has limitations, such as challenges in navigation on uneven terrains, and the absence of a mapping feature to predefine the cleaning area. Upgrading the Raspberry Pi would allow for training more advanced machine learning models, improving detection accuracy and operational speed. Future enhancements could include, improved mobility, cloud-based monitoring for better analytics, and implementing a mapping system to predefine cleaning areas for more efficient operation. The Solar-Powered Autonomous Waste Sorting Robot marks a significant advancement in sustainable beach cleaning, offering an eco-friendly, automated alternative to traditional methods. This project contributes to cleaner beaches and the preservation of marine ecosystems, promoting long-term environmental sustainability.

6.1 Future Scope

- Enhance detection performance by training optimized deep learning models on real world beach datasets, improving accuracy across varied waste types and lighting conditions.
- Incorporate advanced navigation technologies such as GPS, LiDAR, and SLAM for precise localization and intelligent path planning in large, unstructured environments.
- Improve the collection mechanism by designing a multi-degree-of-freedom robotic arm with adaptive gripping, enabling handling of diverse waste shapes and sizes.
- Implement intelligent waste segregation using AI to classify waste into recyclable, non-recyclable, and organic categories with higher precision.
- Develop a rugged, weather-resistant design to withstand harsh coastal conditions such as moisture, sand, and corrosion.
- Extend the application to urban cleaning, parks, and industrial environments, increasing the versatility and impact of the system.

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